

I Learned Python With These Projects

Three beginner-friendly Python projects for learning fundamentals through hands-on practice: a **trivia quiz**, a **password generator**, and a **todo list manager**.

· Projects Overview

#	Project	File	Description
1	Python Trivia Quiz	project1.py	Interactive multiple-choice quiz game
2	Password Generator	project2.py	Customizable password creation tool
3	Todo List Manager	project3.py	JSON-based task management CLI

· Getting Started

No dependencies required · just Python 3.10+.

```
python project1.py    # Trivia Quiz
python project2.py    # Password Generator
python project3.py    # Todo List
```

· Project Details

Project 1 · Python Trivia Quiz (project1.py)

What it does: Interactive multiple-choice quiz with scoring.

Logic Flow:

- 1 Questions randomly selected from a pool
- 2 User selects answer (A/B/C/D)
- 3 Answer validated against correct answer
- 4 Score tracked and displayed at end

Key Concepts: Lists, dictionaries, random module, input validation, loops

Project 2 · Password Generator (project2.py)

What it does: Generate secure passwords with customizable options.

Logic Flow:

- 1 User specifies password length
- 2 User toggles: uppercase, digits, special characters
- 3 Character pool assembled based on selections
- 4 Random characters selected from pool
- 5 Generated password displayed

Key Concepts: String module, random module, conditional logic, string concatenation

Project 3 · Todo List Manager (project3.py)

What it does: Persistent todo list with JSON file storage.

Logic Flow:

- 1 Load existing todos from `todo_list.json`
- 2 Menu: Add / View / Mark Complete / Exit
- 3 User actions modify the todo list
- 4 Changes saved to JSON after each action

Key Concepts: File I/O, JSON serialization, while loops, list manipulation

· File Structure

```
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... project1.py           # Trivia Quiz  
... project2.py           # Password Generator  
... project3.py           # Todo List Manager  
... todo_list.json        # Persistent todo storage
```

· License

Educational project · use freely for learning and reference.